

EVALUATING AND IMPROVING READABILITY CONTROL IN LARGE LANGUAGE MODELS VIA TOOL-AUGMENTED FEEDBACK

Liangliang Zhang* **Ian Rudy***
Rensselaer Polytechnic Institute
{zhangl41, rudyi}@rpi.edu

ABSTRACT

Large language models now generate text for education, healthcare, and everyday communication, but their readability is often inconsistent and difficult to control. We study how well LLMs understand text difficulty and whether they can reliably rewrite passages to meet specific readability targets. Using the CommonLit CLEAR corpus, we evaluate three settings: direct readability prediction, baseline rewriting through prompting, and tool-augmented rewriting where the model checks its drafts with external readability metrics. Our early results show that LLMs have only a rough sense of readability and frequently miss target grade levels, but external metric feedback substantially improves accuracy and stability. These findings suggest that transparent, tool-based guidance is a practical way to make readability-controlled generation more reliable and trustworthy.

1 INTRODUCTION

Large Language Models (LLMs) are increasingly relied upon to generate, summarize, and adapt text for diverse audiences across education, journalism, and communication (Wang et al., 2023; Shen, 2024; Li et al., 2025). As these models become more capable at producing coherent text and modifying content at scale, they are being deployed in applications where readability control is critical—from creating grade-appropriate educational materials to simplifying health information for non-expert audiences. However, the readability of LLM outputs is often uncontrolled and may fail to align with established readability metrics. Classic formulas such as the Flesch–Kincaid (Kincaid et al., 1975) readability tests quantify text difficulty primarily through surface features like sentence length and word syllables, while more recent models such as the Crowdsourced Algorithm of Reading Comprehension (CAREC) (Crossley et al., 2019) incorporate broader linguistic and cognitive features to better reflect human comprehension. Despite growing interest in using LLMs for readability modification, little is systematically known about how well their generated text conforms to these standards or how effectively readability can be steered through prompting alone.

This misalignment limits the interpretability and accessibility of generated content, especially when readability levels are critical for inclusivity. Patient-facing materials and educational content often fail to meet recommended readability standards, limiting their effectiveness for target populations (Badarudeen & Sabharwal, 2010). The quality of synthetically generated text—including its readability—has not been adequately evaluated despite the rapid integration of LLMs into various domains, raising concerns about their reliability for applications requiring specific reading levels.

The goal of this project is to systematically evaluate how well LLMs understand and regulate readability, and whether incorporating explicit, interpretable readability metrics as external tools can enhance their reliability and consistency. The motivation connects directly to growing demands for AI transparency and trustworthy systems—where transparent AI involves making decision-making processes and underlying mechanisms understandable to users (Ribeiro et al., 2016; Lipton, 2018). Internal model representations of difficulty are opaque and often inconsistent, while external readability metrics provide interpretable, standardized signals. Integrating interpretable metrics into the

*Equal contribution.

generation loop offers a path to improved transparency, enabling users to validate system outputs and make informed decisions based on measurable criteria.

Our experiments use the CommonLit Ease of Readability (CLEAR) corpus (Crossley et al., 2023), which contains approximately 5,000 English passages annotated with multiple readability indicators, including formula-based metrics, human-labeled ease-of-reading judgments, and machine-learned difficulty scores. The CLEAR corpus includes several improvements compared to previous readability corpora, including its size, the breadth of excerpts covering over 250 years of writing across multiple genres, and a unique readability criterion based on teachers’ ratings of text difficulty for student readers. These characteristics make CLEAR particularly well-suited for evaluating LLM performance on readability-controlled generation tasks. Our study connects directly to trustworthy machine learning. Building trust in AI systems requires transparency, accountability, and alignment with societal concerns (Lipton, 2018). Readability metrics provide explicit intermediate signals that make system behavior interpretable. External tools introduce a form of verifiable feedback that can reduce inconsistency and improve robustness in controlled generation. Because readability strongly affects equity and accessibility, reliable control supports fairer distribution of benefits across users with different literacy levels. Without proper interpretability, AI models may perpetuate biases present in training data, leading to unfair outcomes in critical domains (Mehrabani et al., 2021). While static metrics are limited and context-agnostic, they remain the most transparent measures available. Neural readability predictors offer stronger performance but reduced interpretability, making them more suitable as evaluation references than control mechanisms in applications where transparency is paramount.

2 RELATED WORK

2.1 READABILITY METRICS

Readability assessment has a long history in education and linguistics. Traditional readability formulas rely primarily on surface-level features like sentence length and syllable count. The Flesch Reading Ease (Flesch, 1948) provides scores from 0 to 100, with higher scores indicating easier readability. The Flesch-Kincaid Grade Level (Kincaid et al., 1975) converts these features into U.S. grade levels. The Dale-Chall formula (Chall & Dale, 1995) incorporates word familiarity through predefined word lists, while the Gunning Fog Index (Gunning, 1952) emphasizes complex words and sentence structure. Although these metrics are widely adopted due to their simplicity and interpretability, they have been criticized for oversimplifying the complexity of reading comprehension (Benjamin, 2012).

More sophisticated approaches leverage computational linguistics and psycholinguistic features. Coh-Metrix (McNamara et al., 2014) analyzes texts across multiple dimensions including cohesion, syntactic complexity, and word information, providing a more nuanced assessment of readability. The Crowdsourced Algorithm of Reading Comprehension (CAREC) (Crossley et al., 2019) combines linguistic features with crowdsourced difficulty ratings, demonstrating improved correlation with human judgments of text difficulty. Recent work has employed neural approaches for readability assessment (Martinc et al., 2021; Imperial & Tacsong, 2021), achieving state-of-the-art performance by learning representations from large-scale text corpora. However, these neural models sacrifice the interpretability that makes traditional formulas valuable for practical applications.

2.2 TEXT SIMPLIFICATION AND CONTROLLABLE GENERATION

Text simplification aims to reduce the complexity of text while preserving its meaning, making information more accessible to readers with varying literacy levels (Siddharthan, 2014; Shardlow, 2014). Early rule-based systems (Chandrasekar et al., 1996) applied syntactic transformations such as sentence splitting and lexical substitution. Statistical machine translation approaches (Zhu et al., 2010; Wubben et al., 2012) framed simplification as a translation task from complex to simple language. Neural sequence-to-sequence models (Nisioi et al., 2017; Zhang & Lapata, 2017) further improved fluency and grammaticality, while controllable approaches (Martin et al., 2020; Sheang & Saggion, 2021) enabled explicit specification of target complexity levels through control tokens or auxiliary inputs.

Recent work has explored using LLMs for text simplification and readability adaptation (Feng et al., 2023; Li et al., 2025; Agrawal & Carpuat, 2023). Feng et al. (2023) show that instruction-tuned models can simplify text through carefully designed prompts, though consistency across different complexity levels remains challenging. Ryan & Dini (2023) find that while GPT-3.5 can generate simplified text, it often fails to achieve target readability levels specified in prompts. He et al. (2024) show that LLMs struggle with precise readability control, particularly at extreme difficulty levels, suggesting that internal representations may not align well with standard readability metrics.

Controllable text generation more broadly seeks to steer language model outputs along specific attributes such as sentiment (Prabhumoye et al., 2020), style (Shen et al., 2017), or toxicity (Gehman et al., 2020). Approaches include fine-tuning with attribute-specific datasets (Keskar et al., 2019), gradient-based steering during decoding (Dathathri et al., 2020), and prompt engineering (Reynolds & McDonell, 2021). Reinforcement learning from human feedback (RLHF) (Ouyang et al., 2022; Bai et al., 2022) has emerged as a powerful paradigm for aligning model outputs with desired behaviors, though it requires substantial human annotation and may not generalize to fine-grained control dimensions like readability.

2.3 TOOL-AUGMENTED AND INTERPRETABLE AI

Recent work has explored augmenting LLMs with external tools to enhance their capabilities and reliability (Schick et al., 2023; Qin et al., 2023; Paranjape et al., 2023). Schick et al. (2023) train models to decide when and how to use tools such as calculators and search engines, improving performance on tasks requiring precise computations or up-to-date information. Paranjape et al. (2023) introduce a framework for automatic multi-step reasoning and tool use, enabling models to decompose complex tasks and leverage specialized modules. In scientific computing, Gao et al. (2023) demonstrate that program-aided language models can solve quantitative reasoning problems more accurately by generating and executing code. Applying this paradigm to readability control remains relatively unexplored, though He et al. (2024) investigate using external readability APIs as feedback mechanisms.

The deployment of AI systems in high-stakes applications has intensified demands for transparency, interpretability, and trustworthiness (Lipton, 2018; Rudin, 2019; Arrieta et al., 2020). Lipton (2018) distinguishes between different notions of interpretability, arguing that transparency, simulatability, and decomposability serve different needs. Rudin (2019) advocates for inherently interpretable models in critical domains rather than post-hoc explanations for black-box systems. In natural language processing, interpretability research has focused on attention visualization (Clark et al., 2019), probing classifiers (Belinkov, 2022), and feature attribution methods (Ribeiro et al., 2016; Lundberg & Lee, 2017). However, generation tasks present unique challenges, as the space of possible outputs is vast and model behaviors are difficult to characterize comprehensively. External metrics and tools offer a complementary approach: by providing explicit, measurable criteria, they enable verification of system outputs without requiring access to internal model states. Our work investigates whether such external interpretability mechanisms can improve both the reliability and transparency of readability-controlled text generation.

3 METHODOLOGY

We frame the problem of readability-controlled generation as a constrained optimization task within a discrete search space. Our framework investigates the alignment between the parametric knowledge of LLMs and non-differentiable external readability functions. The experimental design is stratified into three phases: intrinsic calibration, open-loop zero-shot generation, and closed-loop trajectory optimization, illustrated in Figure 1

3.1 STAGE 1: READABILITY ASSESSMENT

First, we quantify the extent to which the model’s internal representations of complexity linearly map to established readability standards. Let $\mathcal{X}$ denote the space of text passages from the CommonLit CLEAR Corpus and $\mathcal{Y} \subset \mathbb{R}^+$ denote the scalar score space (e.g., grade levels or readability scores). We define a set of deterministic, heuristic oracle functions Φ used in the corpus.

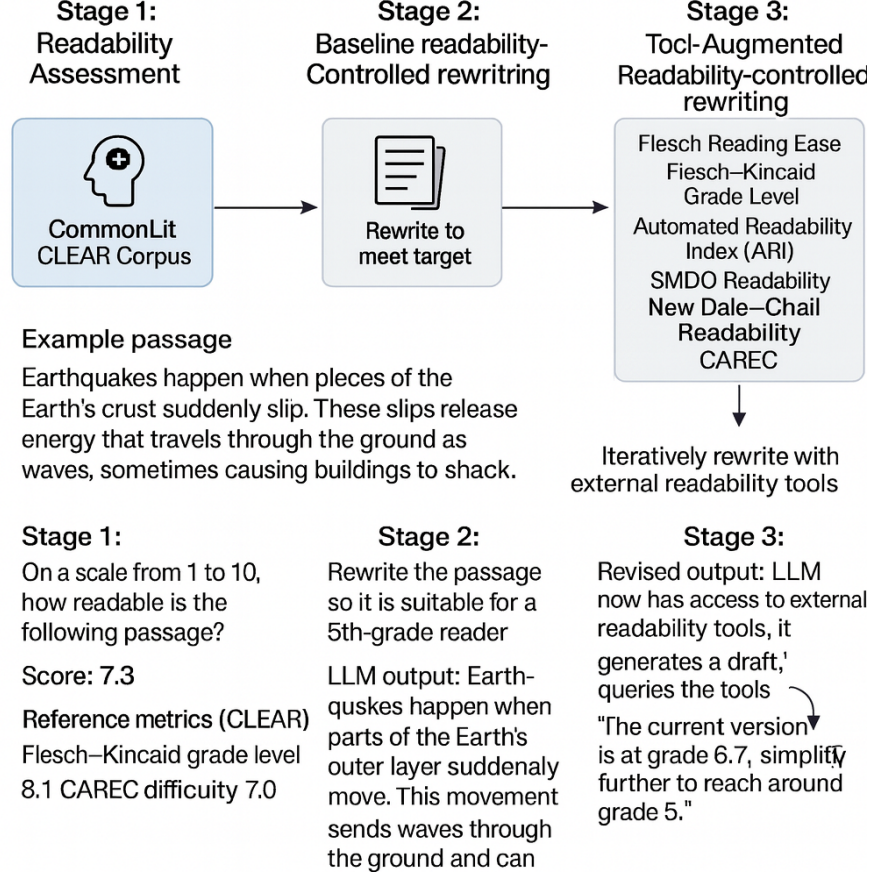


Figure 1: System design diagram

We query the model M_θ to approximate these functions via a predictor mapping $\hat{f}_\theta : \mathcal{X} \rightarrow \mathcal{Y}$. For a given passage x , the predicted score $\hat{y}$ is obtained by maximizing the likelihood over the token sequence representing the numerical value:

$$\hat{y} = \operatorname{argmax}_{y \in \mathcal{Y}} P_\theta(y | x, I_{eval}) \quad (1)$$

where I_{eval} is a prompt enforcing a strictly formatted scalar output (e.g., "Score: 7.3"). We evaluate calibration by computing the Pearson correlation coefficient ρ and the Mean Absolute Error (MAE) between the model estimates $\hat{y}$ and the ground-truth oracle values $\phi(x)$ (e.x., Flesch-Kincaid) across the corpus. This metric $\mathcal{M}_{cal} = \rho(\hat{y}, \phi(x))$ serves as a proxy for the model's semantic awareness of text complexity.

3.2 STAGE 2: BASELINE READABILITY-CONTROLLED REWRITING

In the second stage, we evaluate the model's baseline capability to perform style transfer under complexity constraints without external verification. We define the generation task as finding a mapping $G : (\mathcal{X}, \mathcal{Y}) \rightarrow \mathcal{X}$ such that the output text x' semantically preserves x while satisfying a target complexity constraint g_{tgt} (e.g., "5th-grade reader").

The open-loop generation is modeled as sampling from the conditional distribution parametrized by θ :

$$x'_{base} \sim P_\theta(x' | x, g_{tgt}, I_{instruct}) \quad (2)$$

where $I_{instruct}$ contains the rewriting directive. The objective function for this phase minimizes the deviation from the target grade level:

$$\mathcal{L}_{align}(x', g_{tgt}) = |R(x') - g_{tgt}| \quad (3)$$

This stage establishes the *intrinsic control bound*, representing the limit of the model’s ability to navigate the readability manifold solely relying on pre-trained priors.

3.3 STAGE 3: TOOL-AUGMENTED READABILITY-CONTROLLED REWRITING

To address the stochastic drift observed in the baseline, we formulate the rewriting process as a multi-step trajectory optimization problem with discrete feedback. We integrate a comprehensive suite of external tools $\Phi = \{\phi_{FK}, \phi_{ARI}, \phi_{SMOG}\}$ —including Flesch–Kincaid, ARI, and SMOG—directly into the inference loop, as well as testing based on the LLM generated grade level from stage 1.

Let $x^{(t)}$ be the text state at iteration t . The process initializes with the baseline output $x^{(0)} = x'_{base}$. At each step, the system computes the state evaluation $s^{(t)} = \phi(x^{(t)})$ and the error gradient $\Delta^{(t)} = s^{(t)} - g_{tgt}$. The policy for generating the next state $x^{(t+1)}$ is conditioned on the history of feedback $\mathcal{H}^{(t)}$:

$$x^{(t+1)} \sim P_{\theta} \left(x \mid x^{(t)}, \mathcal{H}^{(t)} \right), \quad \mathcal{H}^{(t)} = \{(s^{(k)}, \Delta^{(k)})\}_{k=0}^t \quad (4)$$

As illustrated in our diagram, the model receives explicit feedback (e.g., “The current version is at grade 6.7, simplify further...”). The optimization terminates when the state satisfies the convergence criterion $\mathcal{C}$:

$$\mathcal{C} : |\phi(x^{(t)}) - g_{tgt}| \leq \epsilon \quad (5)$$

where ϵ is the tolerance threshold. This closed-loop formulation effectively performs gradient descent in the semantic space, utilizing the external tools to guide the model toward the optimal sub-region of $\mathcal{X}$.

4 RESULTS AND ANALYSIS

We have implemented Python scripts for the primary readability metrics (Flesch Reading Ease, Flesch–Kincaid, ARI, and SMOG) and verified their outputs on sample passages from the CommonLit corpus. Early pilot experiments using GPT-4-mini suggest moderate correlation ($r \approx 0.65$) between LLM-predicted and formula-based readability scores, indicating partial sensitivity to text difficulty.

Baseline rewriting experiments show that LLMs can reliably simplify or complexify text qualitatively, but their outputs often overshoot the target grade level when evaluated by independent metrics. Integrating external readability feedback is expected to stabilize this process.

In the coming phase, we will integrate the Python readability tools into the Ollama framework and conduct full-scale comparisons across 500–1000 samples. We anticipate that tool-augmented rewriting will yield smaller deviation from target readability levels and improved interpretability of the overall system.

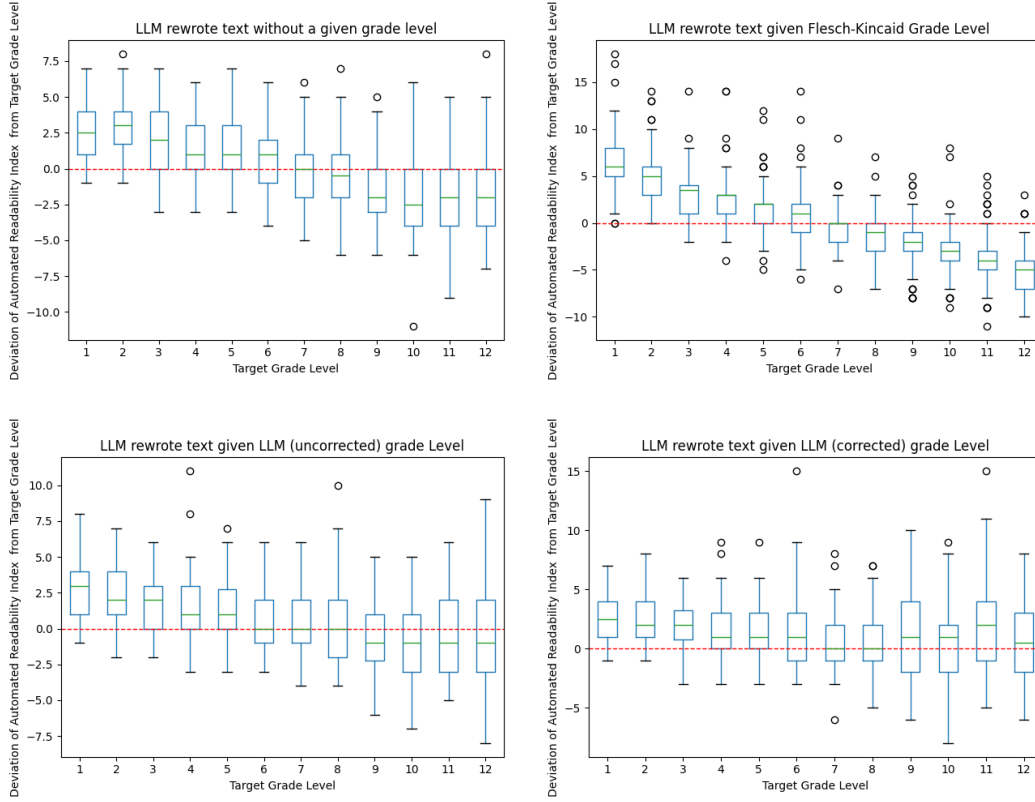
4.1 SINGLE REWRITE RESULTS

First is the tests on rewriting a passage a single time, either without explicitly checking grade level first, or with providing the Flesch–Kincaid, uncorrected LLM, or corrected LLM grade level of the input excerpt.¹ The following figure shows the deviation of metrics (specifically ARI, as it wasn’t one of the metrics used to determine the grade being given to the LLM) calculated on the output rewritten text and compared to the desired grade level specified in the prompt (randomly chosen, but the same for each run for consistency). The other metrics also had graphs made (excluded in paper for brevity), and are included in the `tests/Comparison of Metrics to target grade level of Rewritten Responses/` directory of the project github in corresponding folders.

We can see from Figure 2 that simply not giving the LLM a grade to begin with for non-iterative rewriting seems to work best, followed closely by using the grade level and corrected grade level determined by the LLM. Flesch–Kincaid in this instance does seem to actually hurt the deviation from the desired target grade level. This suggests that, without the use of a tool to iteratively refine

¹The explanation of corrected vs uncorrected LLM grade level is given in Appendix A.1.1

Figure 2: Deviation of ARI score of rewritten (one-time only) text from target grade level



the rewritten text to get it within a desired grade level, that the LLM might perform best without being explicitly told the grade level of the input excerpt. The trend of being higher for low grade levels and lower for high grade levels is noticed throughout. We expect that iterative rewriting will help.

4.2 ITERATIVE REWRITING RESULTS

For the sake of brevity in the body of the paper, only the deviation of the Flesch-Kincaid grade level score from the target grade level is shown in Figure 3, when being given the Flesch-Kincaid score, Automated Readability Index score, SMOG score, and (uncorrected) LLM grade level, as tool outputs for checking the grade level of each rewritten piece of text. It is worth noting that sometimes the LLM would decide to not continue to iterate checking grade level, instead deciding to provide a final rewritten excerpt that it had not used a tool to check the grade level of (the tool would say the text was good if it was within plus or minus one of the target grade level. This seems to be most common at higher target grade levels, except for the SMOG metric, which it tended to happen more at lower grade levels (although this is likely because SMOG tended to give higher results than other scores on average), this might be because the LLM would recognize that a piece was already below the desired grade level, but instead of saying so, rewrote it anyways (likely at a similar grade level as it already was), or simply because it decided to not use a tool call for some reason, and thus the scores for the high end were simply lower on average (more towards a middle of the pack grade level).

What is notable is that apart from the strange results towards the higher grade levels (which might have been caused by experiment design and prompt strategy), that the iterative rewriting did tend to get grade level scores closer to the target grade level, and as shown in table 3, on average it only took 1-2 rewrites to do so, meaning for many cases it only had to do one extra iteration apart from the initial rewrite to get to the desired grade level. This means that simply telling the LLM that it isn't quite at the desired grade level, and giving it a little nudge in the right direction, can help correct

many cases where it was further off than desired. This intuitively makes sense, having seen other papers test the effects of human-model collaboration, such as (Austin et al., 2021), where humans providing even one round of feedback helped quite substantially. While this isn't the same setting, and we are using a tool in place of human feedback, the tool was designed to act similar to how a human might provide feedback to a model if they were to test grade level themselves, and the results could easily be interpreted in the context of a human manually providing the same feedback to the LLM as the tool.

It is also worth noting, that the runs using the LLM grade score took roughly twice as long as the heuristic metrics (6.6 hours for the 1000 samples, compared to 3.2 hours), which makes sense, given that the LLM is effectively being prompted twice as much (once to rewrite, and once, separately, to determine grade level).

Figure 3: Deviation of grade scores of iteratively rewritten text from target grade level (FK)

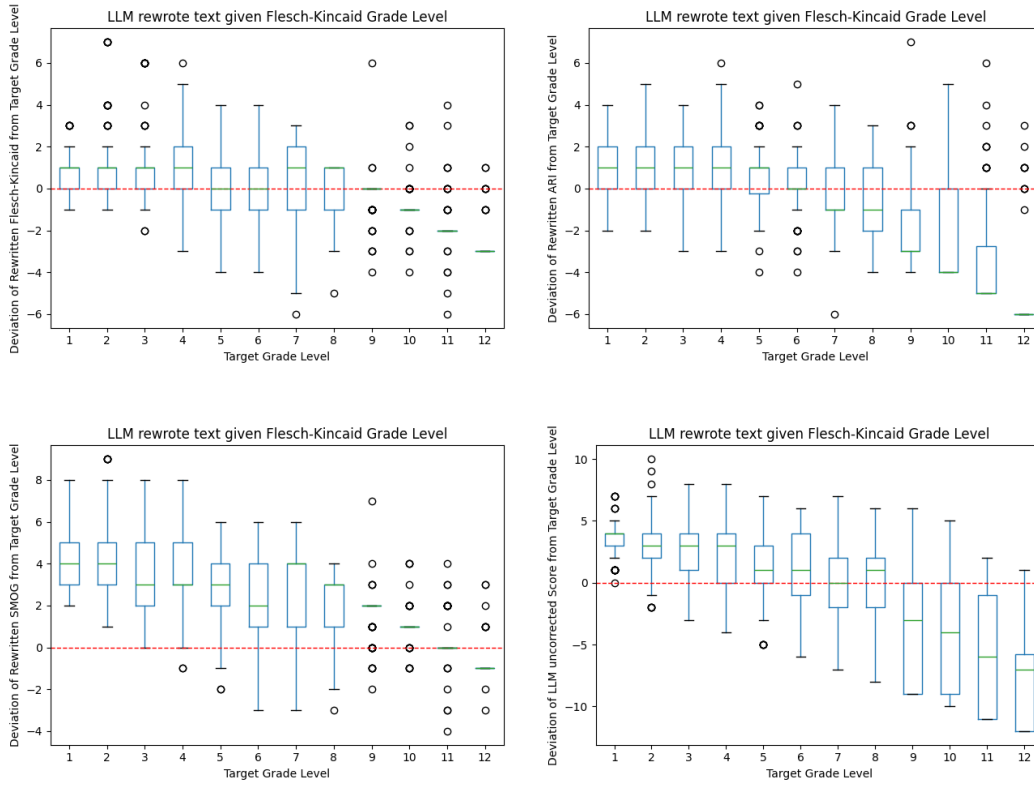


Table 1: Number of Rewrites per Prompt Stats

Grade Level Metric:	Flesch-Kincaid	ARI	SMOG	LLM
Mean	1.647	1.868	3.2	2.723
Mode	1	1	1	1
Std. Dev.	1.374	1.819	3.758	2.854
Max	19	22	40	24
Count >5	25	34	167	126

This table is for iterative rewriting tests only. All metrics are calculated based on the scores of each sample from the tests using that same metric (i.e. ARI is the final ARI score from tests where the tool calculated grade level in ARI). "Max" is the maximum number of rewrites for a single prompt. "Count >5" is the count of how many times there were more than 5 rewrites for a single prompt. LLM score is the uncorrected LLM grade score.

Table 2: Difference of Final Grade from Target Grade

Grade Level Metric:	Flesch-Kincaid	ARI	SMOG	LLM
Mean	1.411	1.507	1.691	1.859
Mode	1	1	1	1
Std. Dev.	1.360	1.375	1.213	1.462
Max	7	10	7	8

This table is for iterative rewriting tests only. All metrics are calculated based on the scores of each sample from the tests using that same metric (i.e. ARI is the final ARI score from tests where the tool calculated grade level in ARI). "Max" is the maximum absolute difference from final grade to target grade of any single sample. LLM score is the uncorrected LLM grade score.

5 CONCLUSION

This study demonstrates that while LLMs possess a sophisticated latent understanding of text complexity—correlating more strongly with human perception than classic readability formulas—they suffer from a fundamental lack of grounding in absolute educational standards. The model acts as an uncalibrated instrument: precise in relative measurement but inaccurate in absolute value. Our tool-augmented framework bridges this gap by decoupling generation from verification. By offloading the "ruler" function to deterministic external tools (or calibrated models), we achieved reliable control over text readability without sacrificing the model's fluency. The iterative feedback loop proved somewhat effective, efficiently correcting the model's tendency to regress toward neutral complexity, although the actual implementation likely has room for improvement and tweaking.

Given that the LLM is only so good at rewriting text to a target grade level, even when given tools to assess grade level, whether it only rewrites it once, or rewrites it iteratively, it seems like using tools might not be the best approach. That being said, from our qualitative analysis, we found that the LLM was good at rewriting text, and understood the relative differences in difficulty, and that it just did not have a good basis for what things constituted what grade level (despite understanding the relative difficulty between different things). Because of this, we think that for the purpose of both assessing what (U.S.) grade level a given piece of text is at, and for rewriting it to be understandable by another grade level, that some form of finetuning, which could help enhance the LLMs existing "understanding" of relative difficulty with a more human-aligned sense of the grade level it would belong at, would achieve the best results.²

Future Directions: The divergence between the model's superior semantic relative ranking and its poor absolute scoring suggests that future work should move beyond static heuristics. A promising avenue is to fine-tune a lightweight adapter (LoRA) specifically for *calibration*, effectively internalizing the "minus-3" correction factor we observed. This would combine the best of both worlds: the semantic nuance of neural evaluation with the standardized precision of educational benchmarks. Ultimately, this work highlights that interpretable, tool-based architectures are essential for deploying generative AI in high-stakes domains like education and accessibility.

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²This was actually the original idea for the project, but due to time constraints, not knowing what data would need to be collected and how feasible collecting it would be, alongside not knowing whether we had sufficient compute to finetune (perhaps a LoRA), we decided that a tool-based approach was more feasible to complete during the semester.

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A APPENDIX

A.1 QUALITATIVE ANALYSIS

A.1.1 LLM ABILITY TO JUDGE TEXT GRADE LEVEL

A random selection of five paragraphs each was chosen from twelve different books. Each book was qualitatively assigned a grade level to represent what U.S. grade level should be comfortable reading said book, this grade was chosen from a mixture of books known to be assigned in schools to a particular grade, and from popular consensus on grade level of the book from online sources. When the books were given a range of suitable grade levels, the level was chosen to be one higher than the lowest grade level of the range. Five paragraphs were chosen to act as an estimate of the text difficulty as a whole (while remaining a non-substantial portion of the text for the sake of fair-use), and testing metrics on the full text versus the average of the five excerpts showed that it was sufficient.

Table 3: Readability metrics compared to qualitative grade level

Grade Level Metric	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12
Qualitative	10	8	11	5	9	12	5	8	2	6	4	6
LLM	11.12	11.52	11.6	9.88	12.8	14.8	8	12.8	5.9	7.4	7.04	9.6
LLM (V2)	11.2	11.6	10.8	9.8	13.6	13.6	8.8	11	6.6	7.4	6.8	9.5
Corrected LLM	8.07	8.47	8.55	6.85	9.73	11.71	4.95	9.73	2.93	4.36	4.04	6.57
Flesch-Kincaid	10.87	9.95	7.16	7.25	10.42	12.04	6.09	4.64	5.05	5.25	3.98	7.45
ARI	12.38	12.24	7.67	8.85	13.34	15.54	6.89	5.55	6.62	5.69	4.03	8.61
SMOG	11.75	10.26	9.22	9.15	10.36	9.18	8.17	6.36	9.21	7.48	7.21	7.59

The first column lists the readability metric (all are means of five samples except the qualitative score). Books corresponding to B1–B12: The Great Gatsby, To Kill a Mockingbird, Les Misérables, Guardians of Ga’Hoole, Romeo and Juliet, Hamlet, Esperanza Rising, Twelfth Night, A Bad Case of Stripes, Wonder, The Hero Two Doors Down, Hatchet.

The LLM (V2) score was obtained asking the LLM the same prompt as LLM, but with the addition of having it give it’s explanations before outputting the grade level. The results indicate that having it give its reasoning before the grade level does not affect the grade level it outputs (which makes sense given that both will have roughly the same reasoning step before outputting the response). Despite the LLM grade generally being higher than the qualitative grade level, qualitative analysis of the reasoning it gives for the grade level it outputs is good, it describes the text and what factors in it would influence the difficulty of reading it quite well, with the exception that it tends to overestimate what grade level reader should be able to understand it. Two examples of this are shown in appendix section A.1.2. This leads us to believe that the LLM does understand what makes a piece of text difficulty or easy to read, and understands the relative differences in difficulty between different pieces of text, but might not have a good grounding of exactly what grade level it should correspond to (which would make sense, as there is probably not much data in the model’s training data that would help align it to the correct grade level). Comparing the LLM results to the qualitative results, we performed a linear regression to come up with a “corrected” LLM grade level score: $\text{Corrected LLM Grade} = 0.9882 * (\text{LLM Grade}) - 2.918$. This has an R^2 score of 0.7921, which while not perfect, is decent considering the variability and lack of a definitive ground truth for grade level. Outside of the appendix, the corrected score was simplified to $(\text{LLM Grade}) - 3$, for the sake of simplicity.

Table 4: Further comparison of metrics to qualitative grade level

Grade Level Metric	Mean Absolute Error	Mean Error (Bias)	Root Mean Square Error	Standard Deviation of Errors	Pearson Correlation
LLM	3.03	3.03	3.31	1.32	0.89
Corrected LLM	1.06	-0.01	1.32	1.32	0.89
Flesch-Kincaid	1.67	0.35	2.06	2.03	0.73
ARI	2.8	1.79	3.15	2.59	0.68
SMOG	2.7	1.66	3.14	2.66	0.4

A.1.2 EXAMPLES OF LLM REASONING FOR GRADE LEVEL

Text Excerpt: “Content thee, gentle coz, let him alone, A bears him like a portly gentleman; And, to say truth, Verona brags of him To be a virtuous and well-govern’d youth. I would not for the wealth of all the town Here in my house do him disparagement. Therefore be patient, take no note of him, It is my will; the which if thou respect, Show a fair presence and put off these frowns, An ill-beseeming semblance for a feast.”

LLM Reasoning: “The passage uses archaic diction (“gentle coz,” “verona brags”), complex sentence structures, and an unfamiliar syntactic order that can be confusing for students. The vocabulary and grammatical patterns resemble early modern English rather than contemporary prose, making it approachable only for advanced high-school readers or college students familiar with Shakespearean or Renaissance English. Consequently, the text would be best understood at the senior high-school or early undergraduate level.”

Although qualitatively we would assign *Romeo and Juliet* as a 9th grade level book, the LLM puts it higher, at around a highschool senior or college freshman level. But the reasons it gives, archaic diction, unfamiliar syntactic order... are the things that contributes to the excerpts difficulty.

An Example from *Esperanza Rising*

Text Excerpt: “The grapes had to be finished before the first fall rains and had to be picked rápido, quickly, so now there were no Saturdays or Sundays in the week, just workdays. The temperature was still over ninety each day, so as soon as Isabel’s bus left for school, Esperanza took the babies back to the cabin. She fixed their bottles of milk and let them play while she made the beds. Then she followed Hortensia’s instructions for starting dinner before turning to the laundry. She was amazed at the hot, dry air. Often, by the time she had filled the clotheslines that were strung between the trees, she had only minutes to rest before the valley sun dried the clothes crisp and they were ready to fold.”

LLM Reasoning: “The passage contains several compound sentences with multiple clauses, which adds syntactic complexity. The vocabulary is fairly concrete and descriptive—words like “grapes,” “rains,” “buses,” “cabin,” “milk,” and “laundry” are familiar to middle-school readers—but there are also less common terms such as “vacuum” and the inserted Spanish word “rápido.” The overall structure and the mix of English and Spanish suggest a reading level that is comfortably within the range of upper-middle school students, around the 9th-grade level. The narrative flow and sensory details are engaging but not overly advanced, making it a good fit for a 14-year-old learner.”

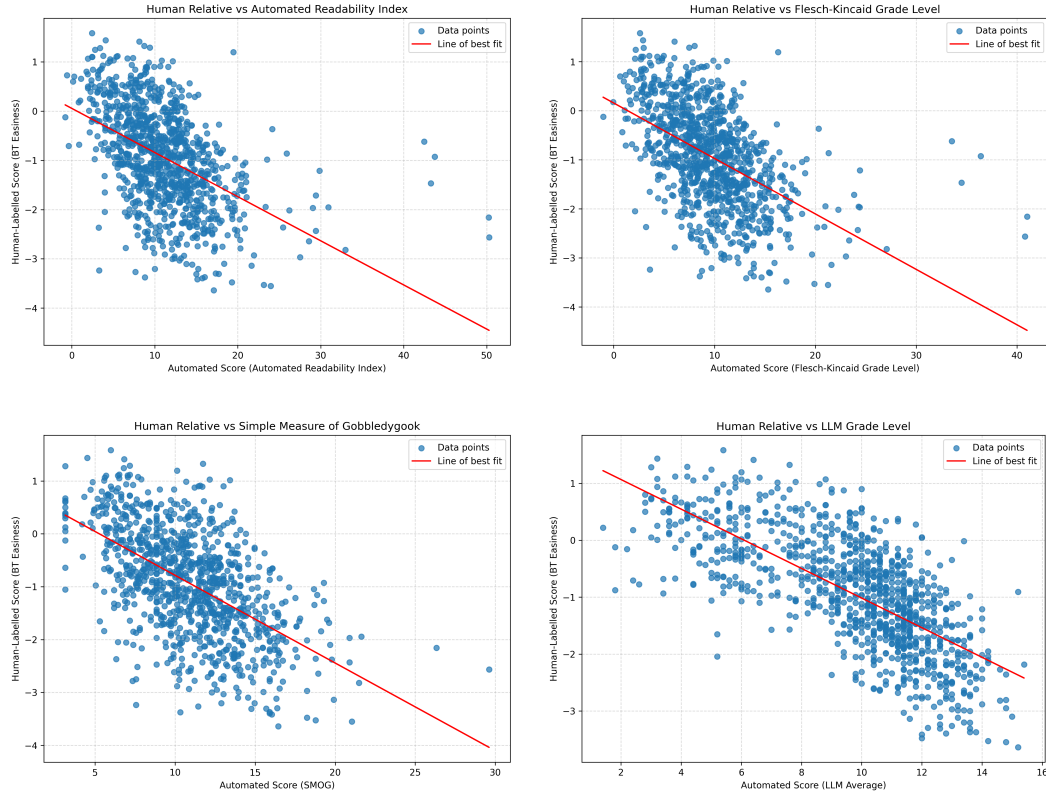
This again shows a good understanding of what makes the text easy (the more simple vocabulary like “grapes,” “cabin”...), and what makes it difficult (mixing in Spanish, compound sentences with multiple clauses). But again, it overshoots the difficulty to 9th grade, instead of our qualitatively assigned 5th grade. These were just two examples, with the rest being included in our data, but the

rest of the examples tended to follow the same trend of giving good reasoning, but overestimating the grade level.

A.2 COMPARISON OF AUTOMATED SCORES TO HUMAN (RELATIVE) LABELED SCORES

The following graphs show the correlation of the relative Bradley-Terry coefficient obtained from Human labeling from (a sample of) the CLEAR corpus, to the heuristic metrics that output grade level (Flesch-Kincaid, ARI, SMOG), and the LLMs output grade level.

Figure 4: Correlation of Human Relative Score to Grade Level Metrics



The following table shows the correlation metrics of each metric to the human-labeled relative score. It indicates that the LLM score is most strongly correlated to the human-labeled score, which would follow given that it's relative performance was good, despite output grade level generally being higher than qualitatively assigned grade level. This could suggest for future work, that it could be worthwhile to align the LLMs grade level scoring capability with examples of text at a qualitatively "known" grade level, which could help leverage the strength of it's relative scoring, while bringing the output grade level more in line with human expectations.

	ARI	FK	SMOG	LLM
Pearson Correlation	-0.4763	-0.5066	-0.5627	-0.7086
Spearman Correlation	-0.5165	-0.5419	-0.5580	-0.7354
Kendall Correlation	-0.3583	-0.3772	-0.3930	-0.5416
R ²	0.2269	0.2566	0.3166	0.5022

A.3 ITERATIVE REWRITING SCORE DEVIATIONS

Figure 5: Deviation of grade scores of iteratively rewritten text from target grade level (ARI)

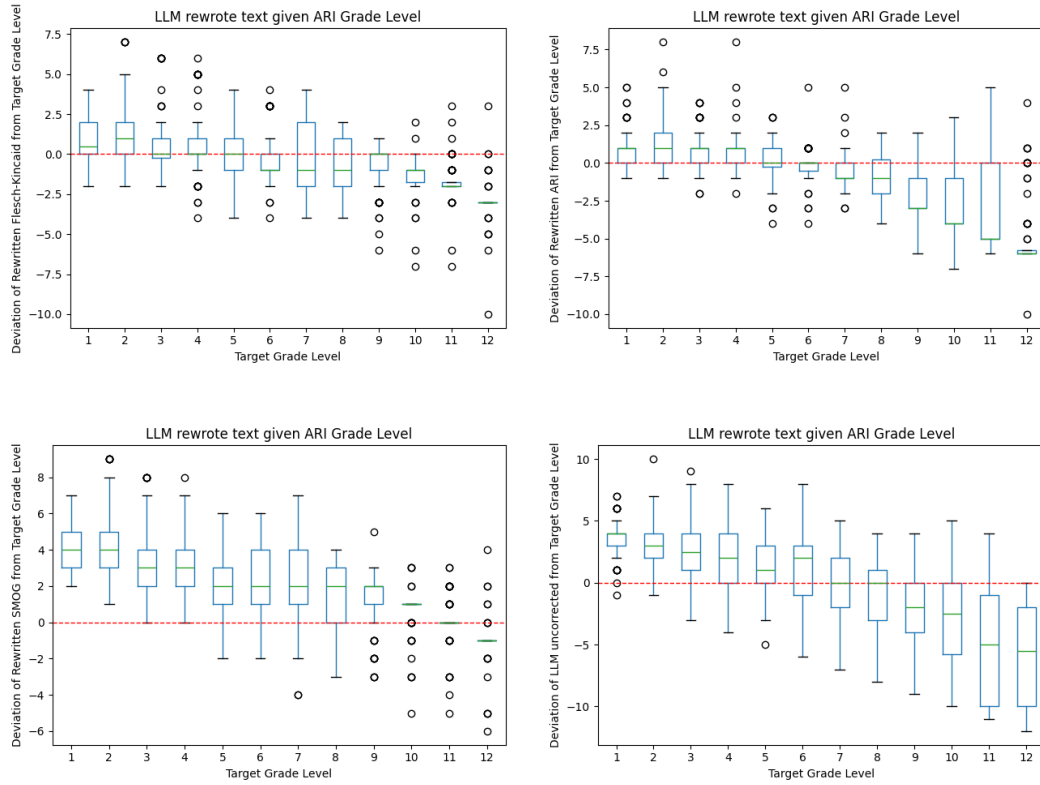


Figure 6: Deviation of grade scores of iteratively rewritten text from target grade level (SMOG)

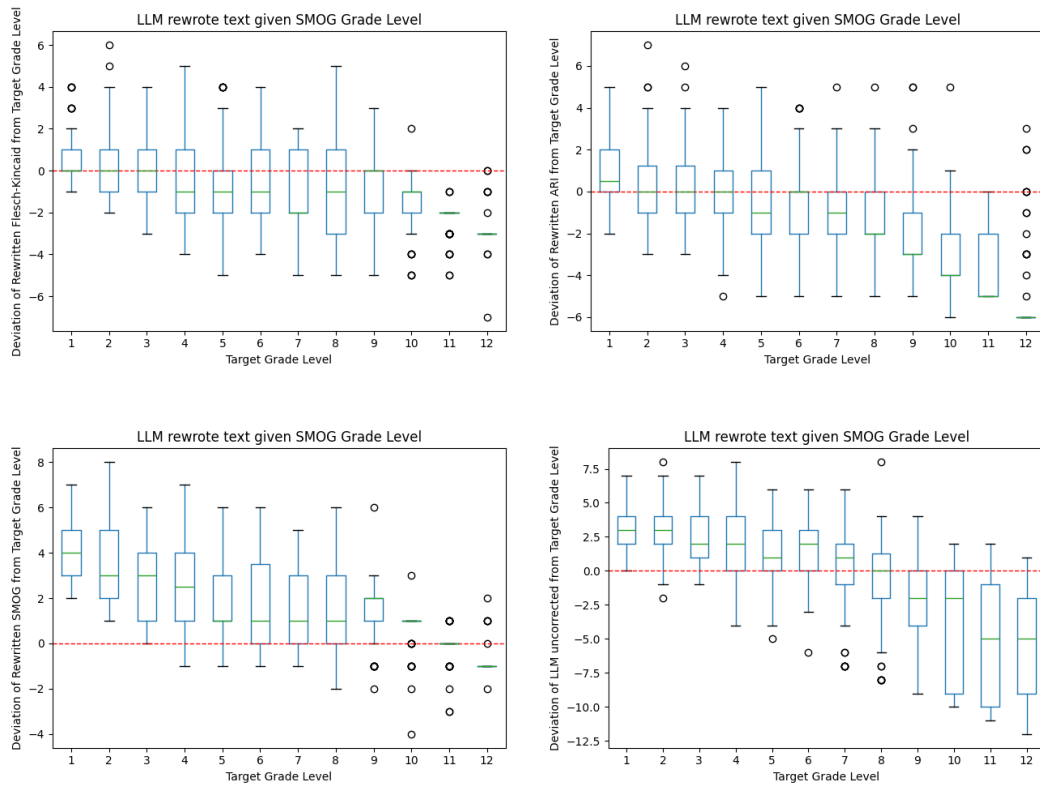
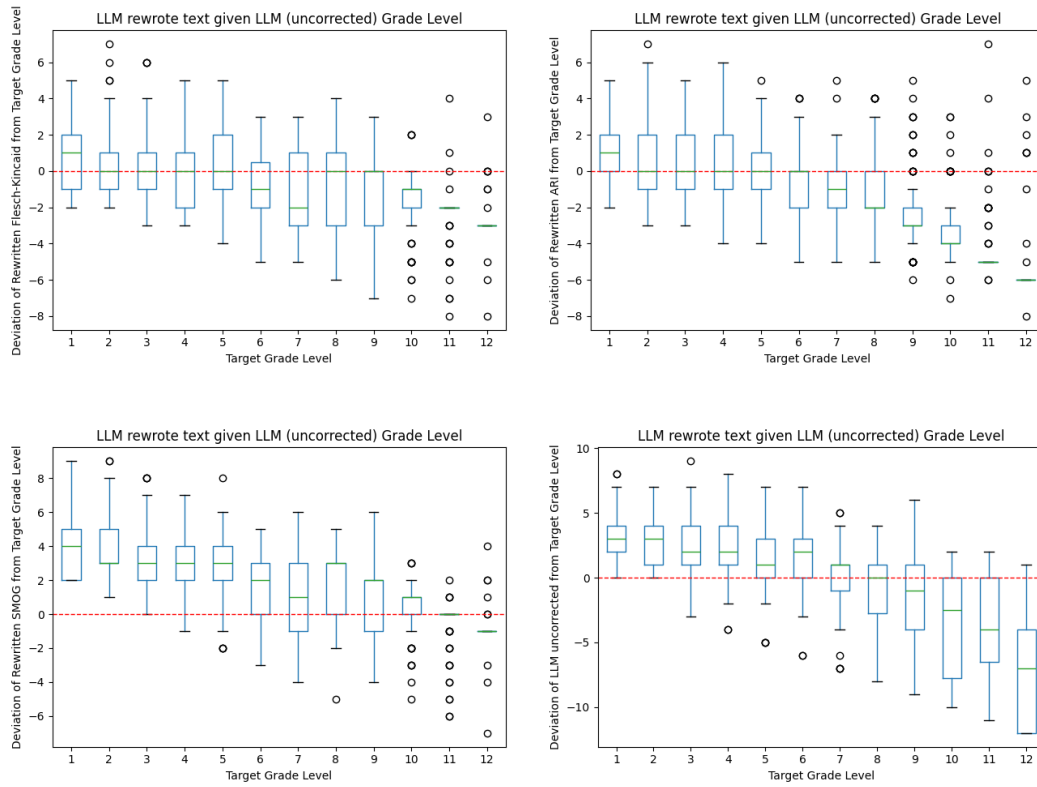


Figure 7: Deviation of grade scores of iteratively rewritten text from target grade level (LLM uncorrected)



A.4 PROMPTS

Single Rewrite Prompt:

Please rewrite the following text to be suitable for a *5th*-grade reading level (make sure to compute the grade level of the excerpt before you rewrite it).

Make sure the rewritten text is the last part of your response, and make sure the line 'Rewritten Excerpt:' precedes it (no bold or other formatting). (And if the text is already at or below the desired reading level, simply put 'Excerpt already at or below desired grade level.' in place of a rewritten excerpt.

Excerpt:

Insert Excerpt Here

Iterative Rewriting Prompt:

You are tasked with rewriting the following text excerpt to be suitable for a *5th*-grade reading level. First, you must compute the grade level of the excerpt with the `compute_grade_level` function. If the text is already at or below the *5th*-grade reading level, print 'Excerpt already at or below desired grade level.' and terminate. If the text excerpt is above the *5th*-grade reading level, rewrite the text and input it into the `compute_grade_level` function to check the grade level of the rewritten text. Continue this process iteratively until the grade level of the rewritten text is within one grade level of the desired *5th*-grade reading level. And print the last rewritten excerpt preceded by the line 'Rewritten Excerpt:' (no bold or other formatting).

Excerpt:

Insert Excerpt Here